Notes: de Silva & Thesmar (RFS, 2024)

Noise in Expectations: Evidence from Analyst Forecasts

Contents

1 Big picture

- **Question.** How much of subjective forecasting error comes from private/soft information, bias, and pure noise?
- **Setting.** The paper studies sell-side analyst forecasts of firm earnings per share from I/B/E/S, matched to Compustat and CRSP, over 1989–2021.
- **Key contrast.** Analysts beat machine/econometric forecasts at short horizons but lose to them at long horizons.
- **Main decomposition.** The paper decomposes the performance gap between analyst forecasts and econometric forecasts into: soft information, public-information bias, soft-information bias, and noise.
- **Central finding.** Analyst soft-information advantage decays quickly with horizon, while analyst bias and noise rise strongly with horizon.
- **Noise matters.** Long-horizon analyst forecasts are not only biased; they also contain forecast variation unrelated to information or realizations.
- **Implication for CG regressions.** Rising noise can mechanically make the Coibion–Gorodnichenko error-revision coefficient fall and even turn negative, independently of true overreaction.
- **Forecast combination result.** Adding analyst forecasts to econometric forecasts helps at long horizons, but the gain is modest because extracting soft information also imports noise.
- **Model contribution.** A parsimonious bounded-rationality model with a noisy cognitive default matches the joint term structures of bias and noise.
- **Economic takeaway.** Subjective forecasts are valuable near term because analysts possess soft information, but long-term forecasts are contaminated by increasing bias and cognitive/noisy-default errors.

2 Data and measurement

2.1 Forecast sample

- The authors collect reported fiscal year-end EPS and announcement dates from the I/B/E/S actuals file for US firms with announcements from 1989 to 2021.
- For each fiscal year t, they collect analyst EPS forecasts issued within 45 calendar days after the annual report release date.
- This timing choice reduces differences in analysts' observable information sets and avoids mechanically stale forecasts.
- The forecasting horizons are
$$h \in \{0.25, 0.5, 0.75, 1^*, 1, 2, 3, 4\},$$
where the first four are quarterly horizons and the last four are annual horizons.
- Forecasts and realizations are normalized by the CRSP stock price at the fiscal year-end:
$$\pi_{i,t+h} = \frac{EPS_{i,t+h}}{P_{i,t}}, \qquad F_t^j \pi_{i,t+h} = \frac{F_t^j EPS_{i,t+h}}{P_{i,t}}.$$
- The consensus forecast is the equally weighted average of individual analyst forecasts for the firm-year-horizon.

Interpretation: The paper studies earnings-to-price forecasts rather than raw EPS levels. This makes forecast errors more comparable across firms and reduces the influence of scale and outliers.

2.2 Econometric forecasts

The econometric forecast approximates the conditional expectation of future earnings using observable variables X_{it}:

$$F_t^e \pi_{i,t+h} \approx \mathbb{E}(\pi_{i,t+h} \mid X_{it}). \tag{1}$$

The paper uses four benchmark or machine-learning methods:

- **Random walk:** current EPS-to-price predicts future EPS-to-price.
- **Elastic net:** penalized linear prediction with cross-validated penalty.
- **Random forest:** tree-based ensemble that averages many decorrelated trees.
- **Gradient-boosted trees:** sequentially fitted shallow trees that improve prediction of residuals.

Interpretation: The econometrician's information set is large but public. The analyst may have additional soft information, such as firm-specific judgment, management communication, industry context, or qualitative interpretation.

2.3 Forecast accuracy measure

For each forecast type, the paper reports normalized mean squared error:

$$\mathrm{MSE}_h = \mathbb{E}\left[(F_t \pi_{i,t+h} - \pi_{i,t+h})^2\right], \tag{2}$$

normalized by the mean squared realization of earnings-to-price at that horizon.

Interpretation: This normalization lets the MSE be read as a percentage utility loss relative to perfect foresight in the paper's interpretive model.

2.4 Core empirical objects

The paper separates four components:

- Θ_h: soft information, or information available to analysts but not to the econometrician;
- Δ_h: bias on public information;
- $(1-\alpha_h)^2\Theta_h$: bias on soft information;
- Σ_h: individual expectation noise.

Interpretation: The most important empirical fact is not only that analysts are inaccurate at long horizons, but that the reason changes with horizon: soft information declines while noise and bias increase.

3 Models and empirical specifications

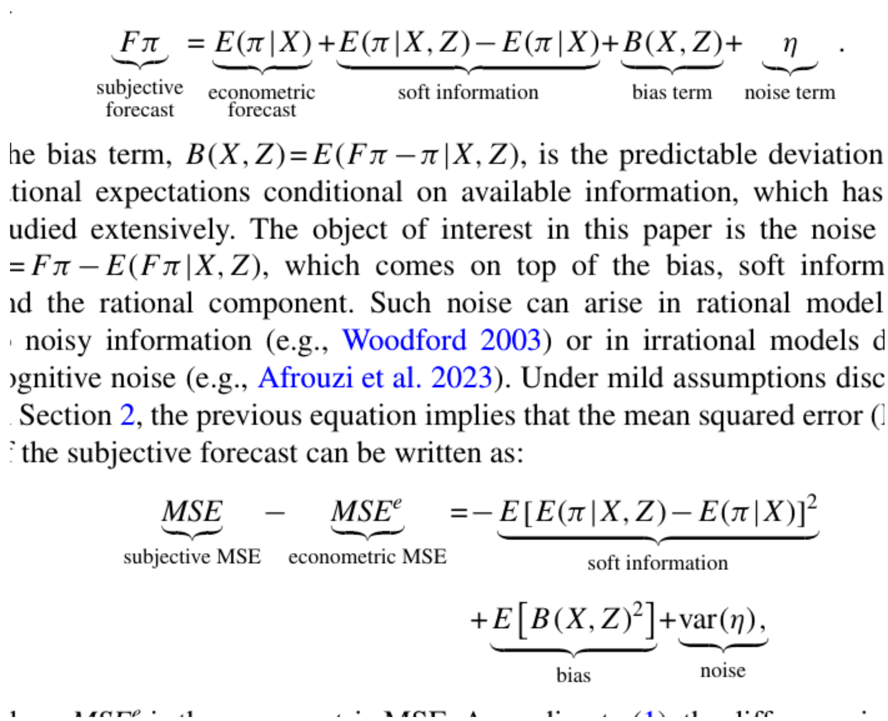
3.1 Forecast decomposition

Let X be the econometrician's public information set and Z be soft information that may be available to analysts. The paper writes a subjective forecast as

$$F\pi = \mathbb{E}(\pi \mid X) + [\mathbb{E}(\pi \mid X, Z) - \mathbb{E}(\pi \mid X)] + B(X, Z) + \eta. \tag{3}$$

- $\mathbb{E}(\pi \mid X)$ is the econometric forecast.
- $\mathbb{E}(\pi \mid X, Z) - \mathbb{E}(\pi \mid X)$ is soft information.
- $B(X, Z)$ is predictable bias.
- η is noise: forecast variation unrelated to information and realization.

Interpretation: This is the conceptual core: subjective forecasts can differ from rational/public forecasts because they contain extra information, systematic bias, or noise.



$$\underbrace{F\pi}_{\substack{\text{subjective}\\ \text{forecast}}} = \underbrace{E(\pi|X)}_{\substack{\text{econometric}\\ \text{forecast}}} + \underbrace{E(\pi|X,Z) - E(\pi|X)}_{\text{soft information}} + \underbrace{B(X,Z)}_{\text{bias term}} + \underbrace{\eta}_{\text{noise term}} .$$

he bias term, $B(X,Z) = E(F\pi - \pi|X,Z)$, is the predictable deviation
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Section 2, the previous equation implies that the mean squared error (
the subjective forecast can be written as:

$$\underbrace{MSE}_{\text{subjective MSE}} - \underbrace{MSE^e}_{\text{econometric MSE}} = -\underbrace{E[E(\pi|X,Z) - E(\pi|X)]^2}_{\text{soft information}} + \underbrace{E\left[B(X,Z)^2\right]}_{\text{bias}} + \underbrace{\text{var}(\eta)}_{\text{noise}},$$

Figure 1: Original paper decomposition of subjective forecasts and MSE.

3.2 Data-generating process and subjective forecasts

The paper decomposes the realization into public information, soft information, and residual news:

$$\pi_i = x_i + z_i + \varepsilon_i, \tag{4}$$

where

$$x_i = \mathbb{E}(\pi_i \mid X_i), \qquad z_i = \mathbb{E}(\pi_i \mid X_i, Z_i) - \mathbb{E}(\pi_i \mid X_i),$$

and $\mathbb{E}(\varepsilon_i \mid x_i, z_i) = 0$.

Individual analyst forecasts are decomposed as

$$F^j \pi_i = x_i + z_i + b_{ij} + \eta_{ij}. \tag{5}$$

Interpretation: The residual η_{ij} is not just analyst disagreement. It is the part of the forecast that cannot be explained by public information, soft information, or predictable bias.

3.3 Consensus forecast and MSE decomposition

For J_i analysts following firm i, the consensus forecast is

$$F\pi_i = \frac{1}{J_i} \sum_{j=1}^{J_i} F^j \pi_i = x_i + z_i + b_i + \eta_i. \tag{6}$$

If forecast noise is uncorrelated with the realization innovation, then

$$\text{MSE}^a - \text{MSE}^e = -\mathbb{E}(z_i^2) + \mathbb{E}(b_i^2) + \text{Var}(\eta_i). \tag{7}$$

Interpretation: Analyst forecasts beat econometric forecasts when soft information is large enough to offset bias and noise. They underperform when bias and noise dominate.

3.4 Identifiable decomposition

Under the paper's assumptions, the MSE gap becomes

$$\mathrm{MSE}^a - \mathrm{MSE}^e = -\Theta + \Delta + (1-\alpha)^2\Theta + \frac{1}{J}\Sigma. \tag{8}$$

- $\Theta = \mathbb{E}(z_i^2)$: amount of soft information.
- $\Delta = \mathbb{E}[(\mathbb{E}(\pi_i \mid X_i) - \mathbb{E}(F\pi_i \mid X_i))^2]$: public-information bias.
- $(1-\alpha)^2\Theta$: bias on soft information.
- $J^{-1}\Sigma$: noise in the consensus, dampened by averaging across analysts.

Interpretation: The consensus forecast averages independent noise across analysts, but it does not average away common bias or common overweighting of soft information.

3.5 Identification of soft information, soft-information bias, and noise

Define residuals after projecting out public information:

$$F_{ij}^* = F^j\pi_i - \mathbb{E}(F^j\pi_i \mid X_i), \qquad \pi_i^* = \pi_i - \mathbb{E}(\pi_i \mid X_i). \tag{9}$$

The paper identifies α, Θ, and Σ from three moments:

$$\mathrm{Cov}(\pi_i^*, F_{ij}^*) = \alpha\Theta, \tag{10}$$

$$\mathrm{Var}(F_{ij}^*) = \alpha^2\Theta + \Sigma, \tag{11}$$

$$\mathrm{Cov}(F_{ij}^*, F_{ik}^* \mid j \neq k) = \alpha^2\Theta. \tag{12}$$

Equivalently,

$$\alpha = \frac{\mathrm{Cov}(F_{ij}^*, F_{ik}^* \mid j \neq k)}{\mathrm{Cov}(\pi_i^*, F_{ij}^*)}, \tag{13}$$

$$\Theta = \frac{\mathrm{Cov}(\pi_i^*, F_{ij}^*)}{\alpha}, \tag{14}$$

$$\Sigma = \mathrm{Var}(F_{ij}^*) - \mathrm{Cov}(F_{ij}^*, F_{ik}^* \mid j \neq k). \tag{15}$$

Interpretation: Cross-analyst covariance identifies common soft information; forecast-realization covariance identifies useful information; residual forecast variance identifies noise.

3.6 Noise and the Coibion–Gorodnichenko coefficient

The CG regression studies the slope of forecast errors on forecast revisions. The paper shows that observed noise mechanically changes this coefficient. If β_{CG} is the observed coefficient and $\bar{\beta}_{CG}$ is the noiseless coefficient, then

$$\beta_{CG} = \bar{\beta}_{CG}\frac{\sigma_{rev}^2 - \Sigma_h}{\sigma_{rev}^2}, \qquad \sigma_{rev}^2 = \bar{\sigma}_{rev}^2 + \Sigma_h + \Sigma_{h+1}. \tag{16}$$

Interpretation: Noise enters forecast errors and revisions with opposite signs. This can push the observed CG coefficient downward even when the underlying noiseless coefficient does not imply overreaction.

3.7 Forecast complementarity

The paper also studies augmented forecasts that combine machine forecasts with analyst forecasts. Under joint normality of soft information and noise, the MSE gain from augmenting the econometric forecast with analyst forecasts can be written as

$$\mathrm{MSE}^a - \mathrm{MSE}^{e+a} = \left[\Delta + (1-\alpha)^2\Theta + (1-\beta)^2\frac{1}{J}\Sigma\right] - (1-\alpha\beta)^2\Theta. \tag{17}$$

Interpretation: Combining forecasts can remove predictable bias and attenuate noise, but it also lowers the weight placed on analysts' soft information. The benefit is largest at long horizons but still limited by noise.

3.8 Noisy cognitive default model

The paper proposes a bounded-rationality model in which forecasters combine a rational forecast with a noisy default:

$$F_t\pi_{t+h} = (1-m_h)d_t^h + m_h\mathbb{E}(\pi_{t+h} \mid X_t, Z_t), \tag{18}$$

where

$$d_t^h = \beta_0 + \beta_x x_t^h + \beta_z z_t^h + \nu_t^h, \qquad \mathrm{Var}(\nu_t^h) = \sigma_v^2. \tag{19}$$

The attention weight is

$$m_h(\kappa) = \frac{\kappa^2}{\kappa^2 + \mathrm{MSE}_h^{FIRE}}. \tag{20}$$

The model implies

$$\Delta_h = (1-m_h)^2\mathbb{E}\left[(\beta_0 + (\beta_x - 1)x_t^h)^2\right], \tag{21}$$

$$\alpha_h = 1 + (1-m_h)(\beta_z - 1), \tag{22}$$

$$\Sigma_h = (1-m_h)^2\sigma_v^2. \tag{23}$$

Interpretation: Long-horizon earnings are harder to predict, so MSE_h^{FIRE} rises, m_h falls, and forecasters rely more on the noisy default. This produces upward-sloping bias and noise.

4 Identification and empirical design

4.1 What identifies noise

- The paper compares forecasts from multiple analysts covering the same firm-year-horizon.
- Common covariance across analysts is attributed to shared soft information or common bias.
- Idiosyncratic residual dispersion that does not covary across analysts is attributed to noise.
- Since analyst noise averages out in the consensus, the contribution of noise to consensus MSE is scaled by $1/J$.

Interpretation: The multi-analyst structure is crucial. Without multiple forecasters for the same realization, it would be hard to distinguish soft information from noise.

4.2 What identifies public-information bias

- The paper estimates $\mathbb{E}(\pi_i \mid X_i)$ and $\mathbb{E}(F\pi_i \mid X_i)$ using machine-learning forecasts based on public observables.
- Public-information bias is the predictable gap between the analyst consensus and the econometric forecast based on public information.
- Elastic net is used for the main decomposition, while tree-based methods provide robustness.

Interpretation: If analysts systematically forecast too optimistically or pessimistically based on public signals, this appears in Δ_h.

4.3 Why timing matters

- Analyst forecasts are taken within 45 days after the annual report release.
- Public variables are constructed from information available at the annual report date.
- Cross-validation for machine learning is done on the training set to avoid look-ahead bias.

Interpretation: The paper tries to align the information timing of analysts and econometric forecasts while keeping forecasts genuinely out of sample.

4.4 Main limitations

- The decomposition relies on assumptions about bias on soft information and uncorrelated noise across analysts.
- The econometrician's public-information proxy depends on the quality of machine-learning estimates.
- Soft information is inferred indirectly, not directly observed.
- Long-horizon forecasts have fewer observations and larger firms, which can affect precision.

Interpretation: The evidence is powerful because it uses multiple analysts and many horizons, but the decomposition is not assumption-free.

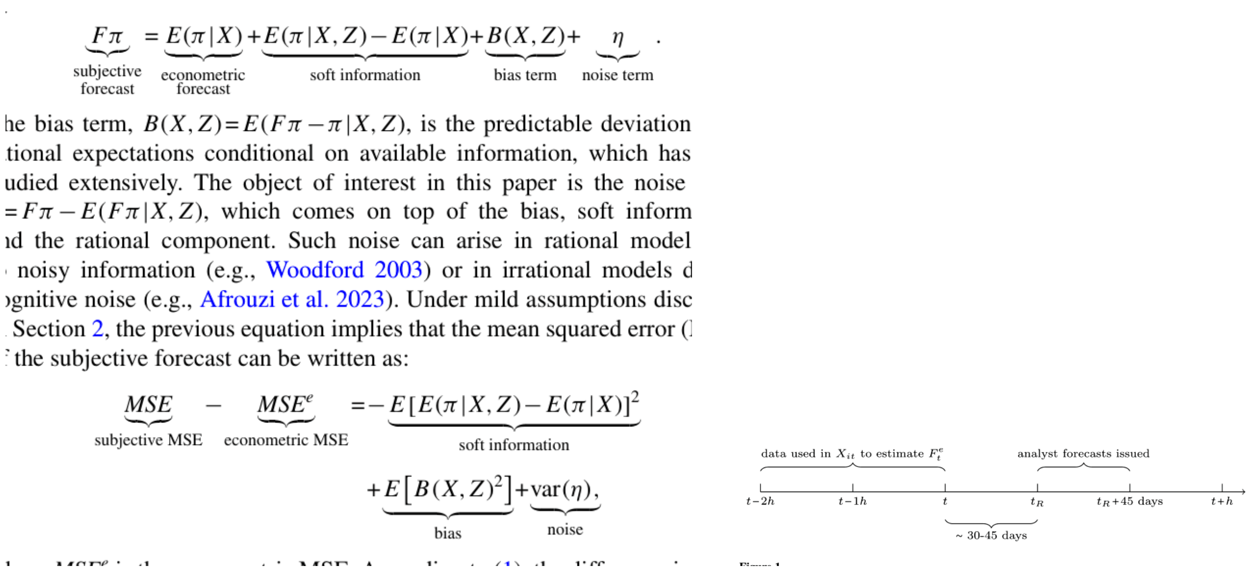
5 Tables and figures

5.1 Key decomposition and Figure 1: what is being measured and when forecasts are formed

Read:

- The decomposition separates subjective forecasts into econometric forecast, soft information, bias, and noise.
- Figure 1 shows the timing: public data used by the econometrician are available before the annual report release; analyst forecasts are collected in the 45 days after the report release.
- Realizations occur at future horizons $t + h$.

Economic message: The timing design makes the comparison between analysts and econometric forecasts meaningful, while the decomposition makes it possible to ask whether analysts are informative, biased, or noisy.



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(a) Forecast and MSE decomposition. **(b)** Timeline of forecast formation.

5.2 Table 1 and Table 2: sample statistics and forecasting accuracy

Read: Table 1:

- Forecast errors become more negative with horizon, meaning forecasts increasingly exceed realizations.
- The mean annual forecast error is about −0.004 at one year, −0.012 at two years, −0.018 at three years, and −0.037 at four years.
- The sample has roughly 4–5 analysts per firm for short and medium horizons, but coverage declines sharply for three- and four-year forecasts.

Read: Table 2:

- Analysts dominate econometric forecasts at short horizons: one-quarter analyst MSE is 4.6%, while the random forest MSE is 15.52%.
- Analyst accuracy deteriorates with horizon: four-year analyst MSE is 46.41%.

- Machine forecasts dominate at long horizons: four-year random forest MSE is 25.76%, far below analyst MSE.

Economic message: Analysts are very useful near term but become poor long-term forecasters. The rest of the paper explains this reversal.

Table 1
Summary statistics

	Count	Mean	SD	10%	25%	50%	75%	90%
A. Analysts' forecasts								
$\pi_{it+h}^{h=0.25} - F_t^j \pi_{it+h}^{h=0.25}$	358,299	0	0.005	−0.004	−0.001	0	0.002	0.005
$\pi_{it+h}^{h=0.5} - F_t^j \pi_{it+h}^{h=0.5}$	311,253	−0	0.008	−0.007	−0.002	0	0.002	0.006
$\pi_{it+h}^{h=0.75} - F_t^j \pi_{it+h}^{h=0.75}$	305,763	−0.001	0.010	−0.010	−0.003	0	0.002	0.006
$\pi_{it+h}^{h=1^*} - F_t^j \pi_{it+h}^{h=1^*}$	305,844	−0.002	0.012	−0.014	−0.004	−0	0.002	0.006
$\pi_{it+h}^{h=1} - F_t^j \pi_{it+h}^{h=1}$	388,895	−0.004	0.031	−0.033	−0.009	0	0.005	0.017
$\pi_{it+h}^{h=2} - F_t^j \pi_{it+h}^{h=2}$	308,682	−0.012	0.052	−0.062	−0.022	−0.003	0.006	0.023
$\pi_{it+h}^{h=3} - F_t^j \pi_{it+h}^{h=3}$	51,722	−0.018	0.072	−0.087	−0.035	−0.006	0.007	0.032
$\pi_{it+h}^{h=4} - F_t^j \pi_{it+h}^{h=4}$	12,816	−0.037	0.110	−0.142	−0.059	−0.013	0.006	0.033
B. Firm-level variables								
$N_{it}^{h=0.25}$	75,970	4.790	4.309	1	2	3	6	10
Total $\text{Assets}_{it}^{h=0.25}$	75,970	9.605	68.532	0.077	0.217	0.834	3.305	12.880
$N_{it}^{h=0.5}$	70,250	4.489	4.166	1	2	3	6	10
Total $\text{Assets}_{it}^{h=0.5}$	70,250	10.151	70.751	0.080	0.229	0.879	3.494	13.744
$N_{it}^{h=0.75}$	68,858	4.489	4.154	1	2	3	6	10
Total $\text{Assets}_{it}^{h=0.75}$	68,858	10.281	71.711	0.081	0.232	0.891	3.546	13.882
$N_{it}^{h=1^*}$	66,983	4.610	4.242	1	2	3	6	10
Total $\text{Assets}_{it}^{h=1^*}$	66,983	10.372	72.149	0.082	0.236	0.917	3.671	14.190
$N_{it}^{h=1}$	75,782	5.189	4.863	1	2	4	7	12
Total $\text{Assets}_{it}^{h=1}$	75,782	9.478	68.437	0.068	0.202	0.794	3.200	12.505
$N_{it}^{h=2}$	64,241	4.839	4.421	1	2	3	6	11
Total $\text{Assets}_{it}^{h=2}$	64,241	10.134	69.989	0.081	0.240	0.924	3.617	13.838
$N_{it}^{h=3}$	20,660	2.518	2.173	1	1	2	3	5
Total $\text{Assets}_{it}^{h=3}$	20,660	20.607	109.421	0.163	0.614	2.470	9.321	32.884
$N_{it}^{h=4}$	7,936	1.630	1.302	1	1	1	2	3
Total $\text{Assets}_{it}^{h=4}$	7,936	23.471	110.206	0.131	0.533	2.964	13.999	44.208

This table shows summary statistics on our final sample, which we construct as described in Section 1.1. In panel A, we show summary statistics for forecast errors at the firm-year-analyst level for our different forecasting

(a) Summary statistics.

Table 2
The term structure of forecasting accuracy: Analyst versus econometrician forecasts

	MSE_h^a	MSE_h^e			
Horizon: h	Analyst	Random Walk	Elastic Net	Random Forest	Boosted Trees
1 Quarters	4.6%	26.1%	20.81%	15.52%	17.25%
		(23.2)	(20.19)	(17.35)	(18.12)
2 Quarters	8.43%	30.0%	19.61%	15.26%	16.91%
		(22.13)	(17.95)	(14.43)	(15.07)
3 Quarters	13.05%	33.87%	22.05%	17.87%	19.28%
		(18.72)	(18.8)	(12.5)	(14.14)
4 Quarters	18.71%	24.92%	25.34%	21.55%	22.45%
		(8.59)	(11.83)	(5.89)	(6.63)
1 Years	9.9%	18.67%	18.63%	15.78%	16.52%
		(12.28)	(15.81)	(12.31)	(12.14)
2 Years	29.19%	34.27%	30.21%	27.05%	29.28%
		(4.03)	(1.04)	(−2.13)	(0.08)
3 Years	33.32%	35.11%	29.11%	26.32%	27.96%
		(1.79)	(−5.09)	(−8.95)	(−6.1)
4 Years	46.41%	37.26%	27.86%	25.76%	29.3%
		(−6.43)	(−14.15)	(−15.7)	(−12.94)

This table contains the mean squared error of analyst forecasts in the first column, denoted as MSE_h^a, and of our econometric forecasts, denoted as MSE_h^e, across different forecasting horizons for forecasts of earnings yields,

(b) Analyst versus econometric forecast accuracy.

5.3 Figures 2–3 and Table 3: term structures of information, bias, and noise

Read: Figure 2:

- Noise Σ_h rises with horizon.
- Soft information Θ_h falls with horizon.
- Public-information bias Δ_h rises sharply at long horizons.
- The weight on soft information, α_h, rises at annual horizons, implying overweighting of soft information.

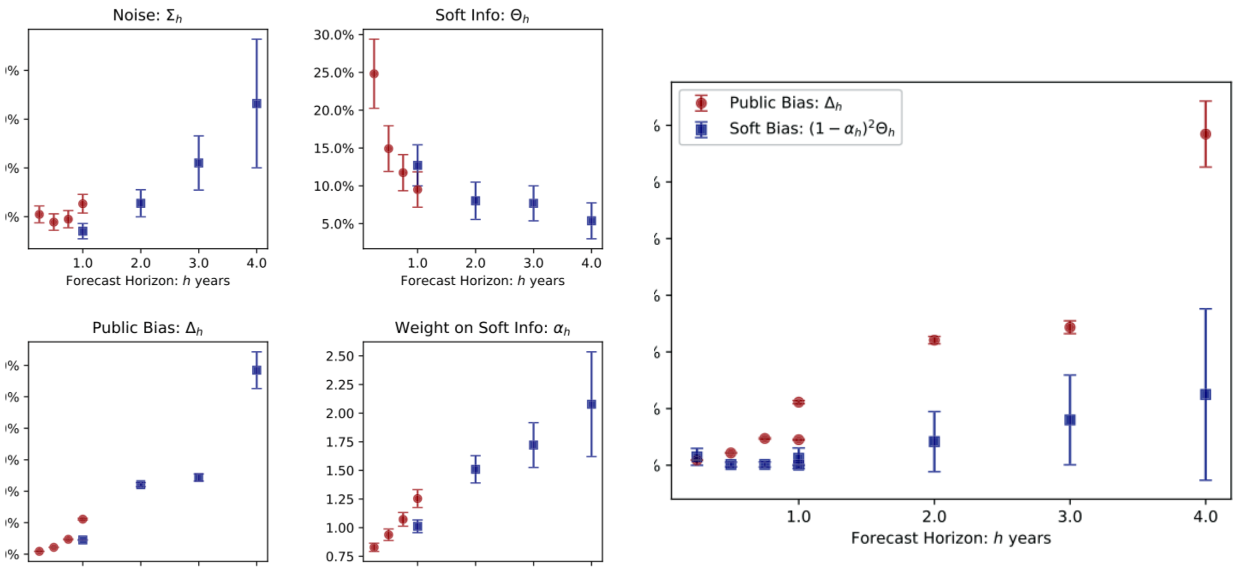
Read: Figure 3:

- Bias on public information becomes much larger than bias on soft information at long horizons.
- Soft-information bias remains modest because the amount of soft information itself declines.

Read: Table 3:

- At one quarter, analysts outperform because soft information contributes −24.82% to the MSE gap.
- At four years, the MSE gap is +40.45%: analysts underperform econometric forecasts.
- The four-year underperformance is driven mainly by public-information bias, 29.23%, and noise, 10.36%.

Economic message: The short-run advantage of analysts is information. The long-run disadvantage is bias plus noise.



(a) Information, bias, and noise. **(b)** Public versus soft-information bias.

Table 3
Decomposition of $MSE^a - MSE^e$

Horizon: h	$MSE^a - MSE^e$	$-\Theta$	Δ	$(1-\alpha)^2\Theta$	$\frac{1}{J}\Sigma$
1 quarter	−22.58%	−24.82%	0.45%	0.73%	1.05%
2 quarters	−12.84%	−14.92%	1.08%	0.06%	0.93%
3 quarters	−8.33%	−11.74%	2.36%	0.06%	0.99%
4 quarters	−2.05%	−9.51%	5.56%	0.61%	1.28%
1 year	−9.81%	−12.71%	2.25%	0.0%	0.64%
2 years	6.32%	−8.02%	11.04%	2.08%	1.23%
3 years	12.57%	−7.69%	12.17%	3.99%	4.1%
4 years	40.45%	−5.38%	29.23%	6.24%	10.36%

This table decomposes the difference between the mean squared error of the consensus forecasts MSE^a, and that of the econometric forecasts, MSE^e, shown in (8) at each forecasting horizon, h. All values are normalized by

Table 1: Decomposition of $\mathrm{MSE}^a - \mathrm{MSE}^e$.

5.4 Figure 4 and Table 4: implications for CG coefficients and forecast combination

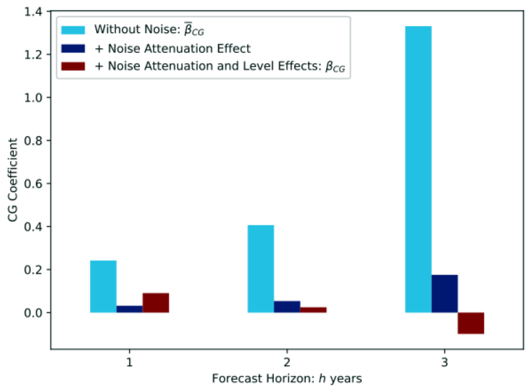
Read: Figure 4:

- The observed CG coefficient declines with horizon and becomes negative at three years.
- The noiseless counterfactual coefficient remains positive and increases with horizon.
- Noise therefore can create an apparent overreaction pattern even when the underlying noiseless bias points in the opposite direction.

Read: Table 4:

- At short horizons, adding analyst forecasts to machine forecasts does not materially improve on the analyst consensus.
- At long horizons, augmented forecasts beat analyst consensus by large amounts.
- At four years, analyst MSE is 46.41%, while random forest plus analyst forecasts achieve 25.29%.

Economic message: Long-term analyst forecasts still contain some soft information, but noise makes that information hard to exploit. Combining forecasts works best when machines can filter analyst bias and noise.



(a) Noise and the CG coefficient.

Table 4
The term structure of forecasting accuracy: Analyst versus econometrician + analyst forecasts

	MSE_h^a	MSE_h^{e+a}		
Horizon: h	Analyst	Elastic Net	Random Forest	Boosted Trees
1 Quarters	4.6%	4.56%	4.81%	4.71%
		(−2.26)	(6.92)	(4.1)
2 Quarters	8.43%	8.49%	8.9%	8.76%
		(1.3)	(6.55)	(5.48)
3 Quarters	13.05%	12.85%	13.02%	12.95%
		(−1.49)	(−0.19)	(−0.67)
4 Quarters	18.71%	17.61%	17.73%	17.69%
		(−4.16)	(−3.14)	(−3.47)
1 Years	9.9%	9.82%	10.01%	10.0%
		(−0.7)	(0.73)	(0.68)
2 Years	29.19%	25.49%	24.13%	25.13%
		(−5.38)	(−6.05)	(−4.63)
3 Years	33.32%	25.1%	24.37%	25.4%
		(−15.11)	(−13.55)	(−11.22)
4 Years	46.41%	26.73%	25.29%	27.55%
		(−16.96)	(−16.82)	(−15.88)

This table contains the mean squared error of analyst forecasts in the first column, denoted as MSE_h^a, and of our

(b) Econometrician plus analyst forecasts.

5.5 Table 5 and Figure 5: noisy cognitive default model

Read: Table 5:

- The estimated default noise is $\sigma_v \approx 0.0635$.
- The key attention parameter is $\kappa \approx 0.0651$.
- Implied attention weights decline with horizon: $m_1 = 0.887$, $m_2 = 0.752$, $m_3 = 0.663$, $m_4 = 0.590$.
- The estimate $\beta_z \approx 2.23$ implies overweighting of soft information in the cognitive default.

Read: Figure 5:

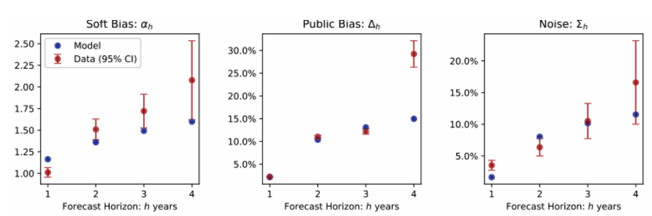
- The model matches the upward slope of soft-information bias, public-information bias, and noise reasonably well.
- The fit is not perfect at the longest horizon, but the model captures the key joint pattern with few parameters.

Economic message: As forecasts become harder, analysts rely more on a noisy and biased default. This links rising bias and rising noise in one mechanism.

Table 5
Minimum distance estimation results

	κ	σ_v	β_0	β_x	β_z
MDE Estimate	0.0651	0.0635	0.0738	0.924	2.2274
Std. Error	(0.0016)	(0.0031)	(0.0029)	(0.0197)	(0.1559)
Implied m_h	$m_1 = 0.887$	$m_2 = 0.752$	$m_3 = 0.663$	$m_4 = 0.590$	

(a) Minimum-distance estimates.



(b) Model versus data.

5.6 Figures 6–7: noise and volatility

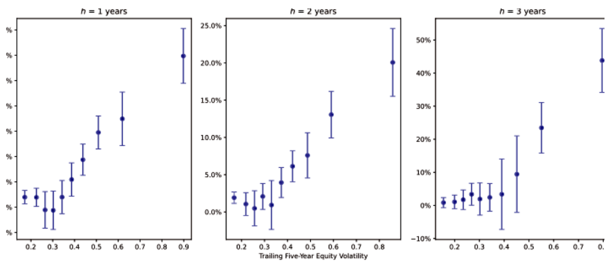
Read: Figure 6:

- Estimated analyst noise rises with equity volatility.
- The relation appears at one-, two-, and three-year horizons.
- High-volatility firms have much noisier long-horizon analyst forecasts.

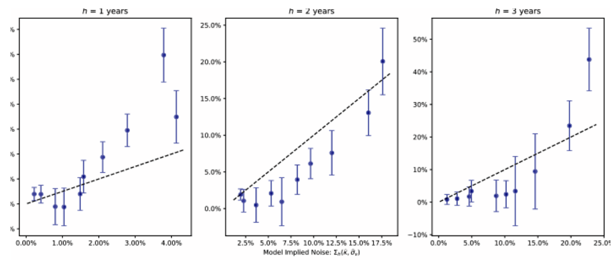
Read: Figure 7:

- The model-implied noise is positively related to empirically estimated noise.
- The model is not a perfect fit, but it predicts the cross-sectional relation between volatility and noise reasonably well.

Economic message: Forecast noise is not randomly distributed across firms. It is higher when the underlying earnings process is more volatile and harder to forecast.



(a) Noise and equity volatility.



(b) Empirical versus model-implied noise.

6 Short summary

- The paper studies analyst forecasts of earnings and compares them to machine/econometric forecasts.
- Analyst forecasts are better at short horizons but worse at long horizons.
- The paper decomposes forecast performance into soft information, public-information bias, soft-information bias, and noise.
- Soft information is large at short horizons but decays rapidly.
- Public-information bias and noise rise strongly with horizon.
- The long-horizon underperformance of analysts is mainly driven by bias and noise, not by absence of econometric sophistication alone.
- Noise changes inference in CG regressions: observed negative long-horizon coefficients can arise mechanically from forecast noise.
- Combining machine and analyst forecasts helps at long horizons, but the improvement over pure machine forecasts is modest because analyst forecasts contain noise.
- A model in which forecasters lean on a noisy cognitive default when the forecasting problem is hard matches the main term-structure facts.
- Noise is higher for high-volatility firms, consistent with the idea that harder forecasting tasks induce more reliance on noisy defaults.

7 Conclusion

7.1 Core conclusion

De Silva and Thesmar show that analyst forecasts are not simply biased forecasts with superior information. They contain a horizon-dependent mix of information, bias, and noise. Analysts provide valuable soft information in the short run, but their long-horizon forecasts become increasingly biased and noisy.

7.2 Economic intuition

At short horizons, analysts can use management guidance, sector knowledge, and recent firm-specific information. This makes them better than public-data-based econometric forecasts. At long horizons, the information advantage fades. Forecasts become harder, and analysts rely more on cognitive defaults that are noisy and biased. This makes the long-horizon analyst forecast a weak signal even when it still contains some soft information.

7.3 Takeaway

The paper's most important lesson is that expectation errors should not be interpreted as bias alone. Noise is quantitatively important and changes how we interpret forecast regressions, forecast combination, and models of belief formation.